CLT and Law of Large Numbers

Strong LLN

- Let $X_1, X_2 \dots X_n$ be i.i.d., mean μ and var σ^2, let $\bar{X}_n = \frac{1}{n} \sum_{j=1}^{n} X_j$ (Sample mean)

- Law of large numbers says that: $\bar{X}_n \to \mu$ as $n \to \infty$ with prob. 1

- This one is also called the strong law of large numbers.

Remembering Markov's and Chebyshev's Inequality

- Law of total expectation says that $E[x]$ can be expressed as a weighted sum of the average of the values below t and the average of the values above t.

- Suppose the bottom 90% of the people make on average of \$30.000 per year, and suppose the top 10% of people make on average of \$230.000 per year.
- We can formulate this as for $X > 0$,

$$E[x] \geq E[X \mid x < t] \cdot P(x < t) + E[X \mid x \geq t] \cdot P(x \geq t)$$

- There is no upper limit for $E[x]$ because $E[x \mid x > t]$ can be, theoretically, infinity. But we can formulate the lower bound of $E[x]$

$E[x] \geq 0 \cdot P(x < t) + t \cdot P(x \geq t)$, and if we move the t to the other side, we get Markov's Inequality

$$P(x \geq t) \leq \frac{E[x]}{t}$$

Let's continue with a real world example:

Let X = income of a random person in US. And let $E[x] = \$50.000$.

So $P(x \geq \$100.000) \leq \frac{E[x]}{\$100.000} = \frac{\$50.000}{\$100.000} = \frac{1}{2}$

But why? Why the probability of selecting a person with \$100.000 income or more cannot be more than 0,5? For example $\frac{2}{3}$?

Let's say we selected 3 random person, two of them have \$100.000 income. What should be the income of the other person for the average to be \$50.000?

It should be -\$50.000 and that's impossible. So for two person to have \$100.000 income, average should be at least \$66.666,666

- Chebyshev's inequality is very similar and derived from Markov's inequality:

$$P(X \geq a) \leq \frac{E[X]}{a} \longrightarrow \text{Let } X \text{ be } (X-\mu)^2 \text{ and } a = (k\sigma)^2$$

$$P((X-\mu)^2 \geq k^2\sigma^2) = P(|X-\mu| \geq k\sigma) \leq \frac{E[(X-\mu)^2]}{k^2\sigma^2} = \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}$$

Weak LLN

- For any $\epsilon > 0$, $P(|\bar{X}_n - \mu| \geq \epsilon) \longrightarrow 0$ as $n \to \infty$

Proof

$$P(|\bar{x}_n - \mu| \geq k\sigma) \leq \frac{E[(\bar{x}_n-\mu)^2]}{k^2\sigma^2} = \frac{Var[\bar{x}_n]}{k^2\sigma^2}$$

$$Var[\bar{x}_n] = Var\left[\frac{1}{n}\sum_{i=1}^{n} x_i\right] = \frac{1}{n^2}\sum_{i=1}^{n} Var(x_i) = \frac{1}{n^2}\overbrace{n \cdot Var(x)}^{\text{because } x_n \text{ is i.i.d}} = \frac{Var(x)}{n}$$

$$\frac{Var[\bar{x}_n]}{k^2\sigma^2} = \frac{Var[X]}{k^2\sigma^2 n} \longrightarrow \lim_{n\to\infty} \frac{Var[X]}{k^2\sigma^2 n} = \lim_{n\to\infty} P(|\bar{x}_n - \mu| \geq k\sigma) = 0$$

or with different notation $P(|\bar{x}_n - \mu|) \geq k\sigma) \to 0$ as $n \to \infty$

- Another way of writing the SLLN is $\bar{x}_n - \mu \to 0$ with prob. 1 (just by doing mid-school algebra)

- But what does the dist. of $\bar{x}_n$ look like?

Central Limit Theorem

- $n^{1/2}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \to N(0,1)$